

Samuel Akinyede, Ph.D.

Lead Applied AI Scientist | AI Systems Architect | Decision Systems Under Uncertainty

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PROFESSIONAL SUMMARY

Applied AI research leader and systems architect with 12+ years spanning technology leadership, enterprise systems, and research-grade machine learning. Turns noisy data, ambiguous policy, and competing operational constraints into auditable models and deployable decision systems across research security, edge sensing, industrial optimization, and geospatial analytics. Ph.D. in Computer Science; leads work from first-principles problem framing and stakeholder co-design through model development, full-stack architecture, validation, and institutional handoff.

TECHNICAL LEADERSHIP & SKILLS

Leadership & Architecture: AI systems architecture, research-to-product delivery, stakeholder co-design, technical roadmaps, model evaluation, uncertainty quantification, auditable decision systems

AI/ML & Decision Science: Python, R, C++, MATLAB, PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, OpenCV, YOLO/DETR, Hugging Face, RAG/agent workflows, SHAP, Bayesian reasoning, Pyomo, SciPy, OR-Tools, NSGA-II, Monte Carlo

Systems & Data: SQL, JavaScript/TypeScript, FastAPI, Flask, Django, Node.js/Express, Streamlit, REST APIs, PostgreSQL/PostGIS, MongoDB, Neo4j, Qdrant, Elasticsearch, Docker, AWS, Azure, GCP

Edge, Geo & Industrial: BLE, Raspberry Pi, Jetson Orin Nano, Arduino/ESP32, GeoPandas, ArcGIS Pro, MLflow, Dagster, techno-economic analysis, SCADA/PLC

EXPERIENCE

Research Consultant & Computer Scientist | Jan 2026 - Present

NSF SECURE Center, University of Missouri-Kansas City

- Own end-to-end product and technical architecture for SENTINEL, translating federal research-security requirements into a multi-tenant intelligence platform that scores proposals across 30+ Critical Technology Areas, generates role/regulation/activity checklists for six institutional roles, and uses five background agents for continuous compliance monitoring.
- Converted FRAME+WORK/RSRMF from an internal workbook into a community-validated operating framework; synthesized 61 practitioner responses into 40 risks across 6 categories, 18 controls, 5x5 scoring, regulatory penalty mappings, an NPRI/clinical overlay, a workshop playbook, and a machine-readable catalog direction.
- Architected and deployed the RIAT prototype on Flask/PostgreSQL/Azure, integrating 276,433 searchable federal screening rows, participant intake, three-level outcome logic, PDF result handoff, admin analytics, and completion-time metrics for CSA validation.
- Designed CARE and a three-layer Risk Management Platform architecture (role x regulation x research activity), combining five contextual dimensions, Bayesian weight-of-evidence reasoning, reference triggers, field baselines, evidence packets, work-level links, and analyst labels to reduce foreign-entity screening false positives.
- Filed a UMKC software disclosure documenting SENTINEL's novel and commercial potential; built the full stack across PostgreSQL, Neo4j, REST APIs, Streamlit, LLM workflows, and auditable decision trails for institutional users.

Graduate Professional Assistant | Jun 2025 - Dec 2025

NSF SECURE Center, University of Missouri-Kansas City

- Led stakeholder discovery, workshops, SME co-design, and qualitative research with research-security professionals; translated findings into product requirements, lifecycle roadmaps, and compliance-focused software interventions.
- Architected MFTRP Hub data/model documentation for a multi-tenant platform spanning 50+ tables, 500+ fields, 90+ relationships, vector and graph collections, Elasticsearch indexes, APIs, business rules, and performance metrics; developed RAG briefings and LLM service concepts for decision support.

Graduate Assistant - Applied AI & Optimization Research | Jun 2024 - May 2025

Missouri Institute for Defense & Energy - Critical Materials Crossroads

- Independently conceived and delivered the Universal Metallurgical Process Simulator, unifying physics-based mass/energy balances, genetic algorithms, NSGA-II Pareto exploration, feasibility validation, and trade-offs across recovery, cost, and environmental impact.
- Built CAPEX/OPEX/NPV/IRR analysis, sensitivity and Monte Carlo uncertainty, LLM explanations, and Flask/REST progress streaming; separately developed Kansas City economic-cluster prediction using federal data, network community detection, XGBoost/SHAP, multi-objective optimization, and conservative calibration.

Graduate Instructor - Computer Science | Sep 2022 - May 2024

Department of Computer Science, University of Missouri-Kansas City

- Taught Discrete Structures, Operating Systems, and Data Structures & Algorithms for incoming M.S. students; designed lectures, assessments, and structured feedback for mixed-background cohorts.

Research Assistant - Edge AI & IoT Systems | Jun 2021 - Aug 2022

Department of Computer Science, University of Missouri-Kansas City

- Validated the ShotTracker UWB sports-tracking platform and engineered the Food OASES web/IoT sensor network with Arduino, ESP32, and environmental probes for data-driven urban agriculture.
- Administered the NSF AISL Moodle/PHP platform on GCP, built a privacy-preserving browser extension, and developed a React Native crowd-safety app with ML inference, a real-time safety index, and in-app visualization.

SELECTED EARLIER TECHNOLOGY LEADERSHIP

Vice President, Information Systems | Sep 2017 - Feb 2019

Tohl Inc., Atlanta, GA

- Directed GIS, asset-management, and industrial-automation programs for public-infrastructure clients in Nigeria, Liberia, and The Gambia; aligned feasibility, environmental, technical, training, and change-management work across client and engineering teams.
- Architected IBM Maximo/ESRI/PostgreSQL spatial systems for Lagos Water Corporation and partnered with Green-flow Engineering to deliver Allen-Bradley/Ignition PLC-SCADA controls supporting MARPOL 73/78 compliance.

Co-Founder / Solution Architect | 2016

Liplus Technology Services, Nigeria

- Co-founded a product studio and led architecture for SI Media, Medic Serve, and Quick Serve; coordinated engineers and data scientists across media, healthcare, and real-time queueing products, with SI Media reaching 500+ Google Play installs.
- Built Django/PostgreSQL and cloud services, OpenCV/Tesseract OCR pipelines, and secure TensorFlow, scikit-learn, and PyTorch prediction workflows for healthcare, queueing, and data-extraction use cases.

Chief Technology Officer | 2014 - 2016

Comrade Air Services, Nigeria

- Set enterprise technology architecture across Cisco networks, Microsoft identity/email, VoIP, backup/recovery, and charter-booking APIs; provided technical leadership to engineering teams across infrastructure and software initiatives.
- Reduced communication costs by 18 percent through a Grandstream VoIP deployment and implemented backup/recovery processes with 99.9 percent reliability to strengthen continuity and protect critical data.

Google Campus Cohort Sprint - Invited Solution Architect | 2016

Shoreditch, London

- Led development of SBFP, an OpenStreetMap-based platform that identified community-verified safe bicycle parking through crowd-sourced safety scoring and real-time geospatial overlays.

Earlier Career: Enterprise Manager, Suburban Broadband West Africa (2013 - 2014) | Systems Analyst, NITDA (2010 - 2011) | Intern, MTN Nigeria (2008)

SELECTED TECHNICAL PORTFOLIO

- Built a geospatial mineral-prospectivity prototype for Pb/Zn exploration using FastAPI, React/MapLibre, PostGIS, MinIO, Dagster, MLflow, probability surfaces, uncertainty layers, and data-support heatmaps; designed an NDA-sensitive RAG diligence assistant for citation-backed Q&A, red-flag extraction, report drafting, and gap analysis.

EDUCATION

Ph.D., Computer Science, University of Missouri-Kansas City. Interdisciplinary focus: Computer Science, Communication Systems & Geosciences. Dissertation: CrossSense: A Quality-Aware AI Framework for Cross-Domain Sensing.

M.Sc., Mobile and Satellite Communication, University of South Wales, United Kingdom. Dissertation: Performance Analysis of Trellis Code Modulation in AWGN and Rayleigh Fading Channels.

B.Tech., Computer Science, Bells University of Technology, Nigeria.

SELECTED PUBLICATIONS & MANUSCRIPTS

- Akinyede, S., Song, S. "AlertBLE: Alert Workzone Hazards Using Hybrid Filtering and Machine-Learning-Enabled BLE," IEEE Internet of Things Journal, vol. 12, no. 22, pp. 46626-46647, 2025. DOI: 10.1109/JIOT.2025.3608718.
- Akinyede, S., Song, S. "Cross-Architecture Compatibility Analysis for Deployable Agricultural IoT Vision Systems," 2025 IEEE Annual Congress on Artificial Intelligence of Things, pp. 107-114, 2025. DOI: 10.1109/AIoT66900.2025.00026.
- Akinyede, S., Song, S. "AIBLE: AI-Powered BLE Localization System for Work Zone Safety Alerts," ICCCN 2025.
- Akinyede, S., Song, S. "Beyond RSSI Thresholds: AI-Assisted BLE Zoning Under Environmental Uncertainty Validated on the NIST Dataset," submitted to the 17th IFIP Wireless and Mobile Networking Conference (WMNC 2026).
- Akinyede, S., Song, S. "Less is More: Deployment-Aware Architecture Selection Outperforms Enhancement Design in Agricultural IoT Object Detection," submitted to IEEE Access.
- Akinyede, S., Caruso, A. "An AI-Driven Framework for Economic Cluster Prediction and Optimization: A Kansas City Metropolitan Statistical Area Case Study," Technological Forecasting and Social Change, submitted/in review.

CERTIFICATIONS, AWARDS & MEMBERSHIP

Project Management Professional (PMP); Certified ScrumMaster; IBM Certified IoT; Social and Behavioral Investigator (IRB); 2025 Community of Scholars; IEEE ComSoc Student Travel Grant; Hack-A-Roo Fall 2024 Graduate Track - 2nd Place; Tech Venture ELEVATIONLAB Scholarship; Member, IEEE.